Git & GitHub-1

Notes: Create a Jira ticket for this assignment, work on the tasks listed below, and submit the task completion document by EOD with clear steps and required evidence (outputs/screenshots)

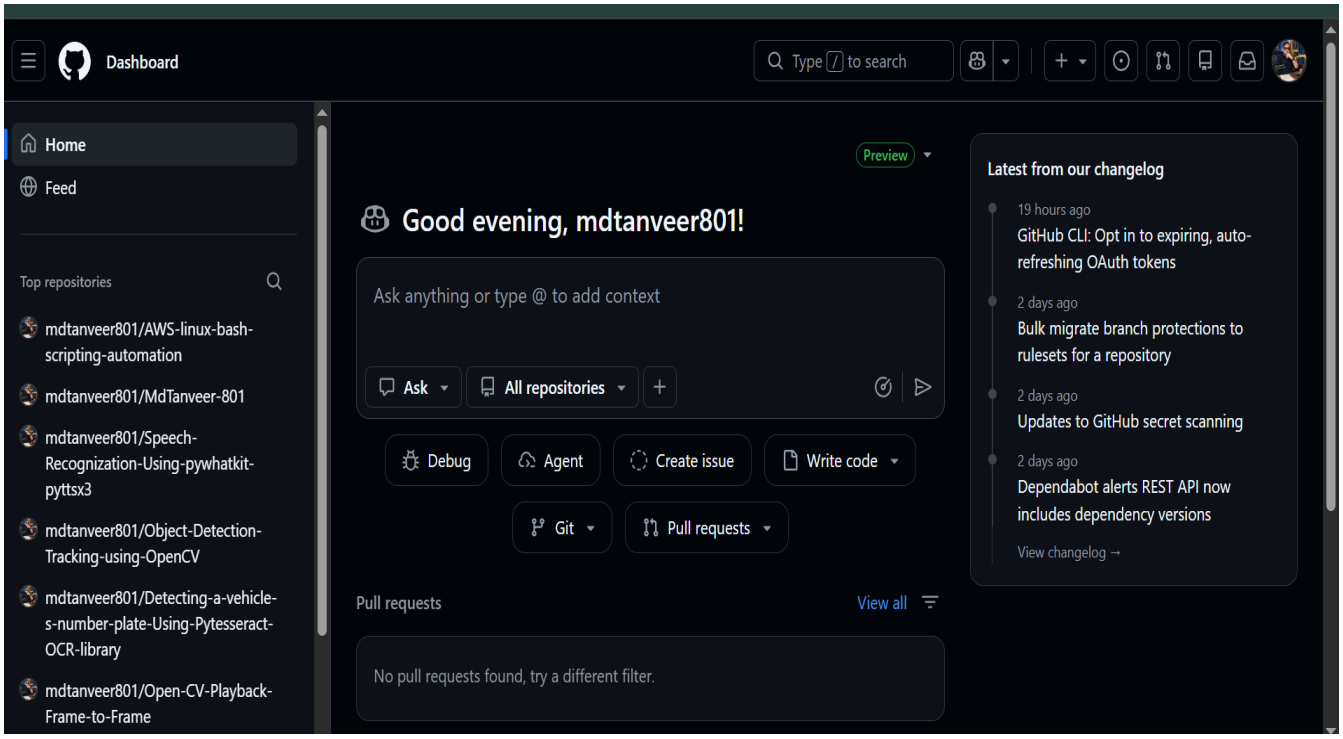
1. Watch the video and write notes..

2. Create a GitHub account..

3. Set up MFA (Multi-Factor Authentication) for the GitHub account using a mobile app..

4. Execute all commands shown in the video.

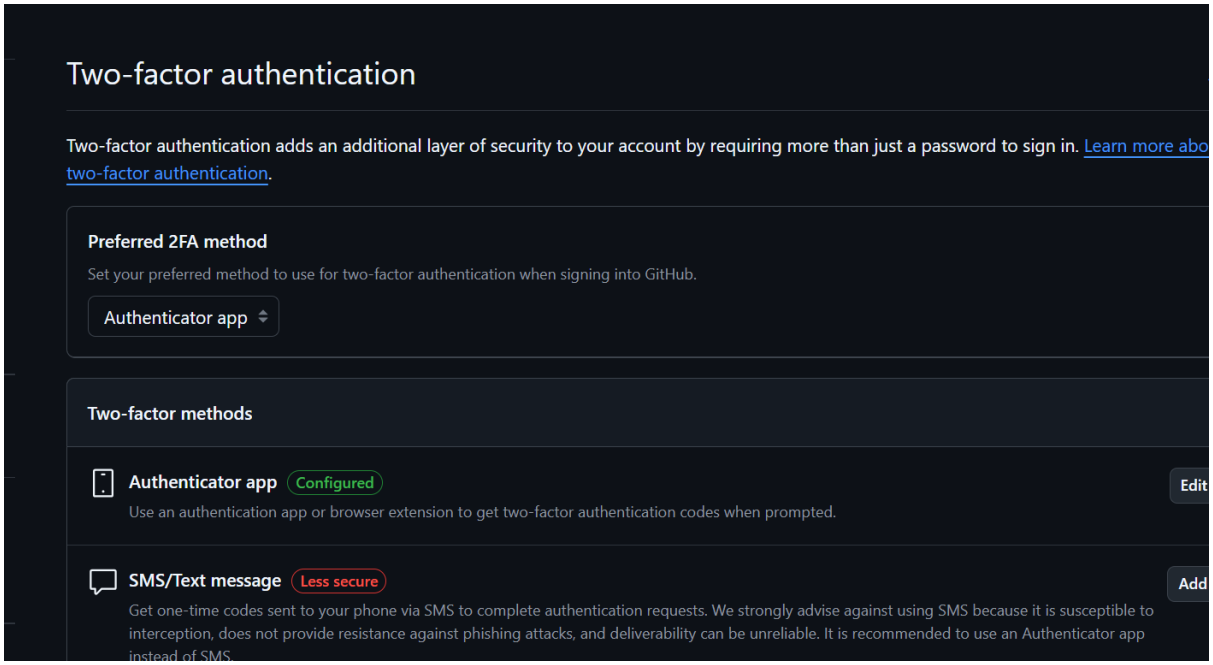
Account has been Created



Git-hub Repositories link:- https://github.com/mdtanveer801?tab=repositories

3. Set up MFA (Multi-Factor Authentication) for the GitHub account using a mobile app..

MFA Done With Microsoft Authentication.



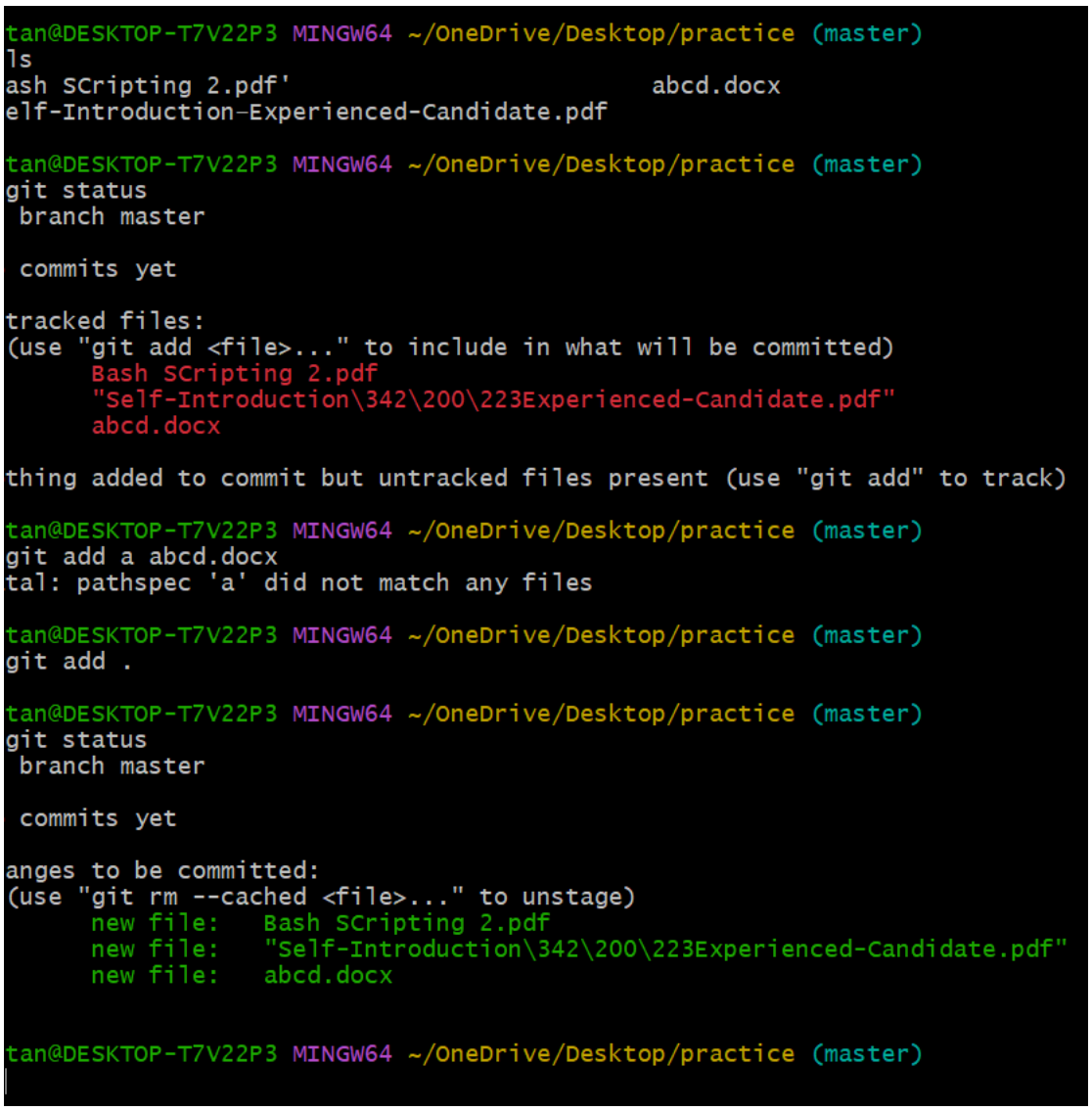
Execute all commands shown in the video.

```
mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (master)
$ ./ ../practice
bash: ./: Is a directory

mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager-core
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
user.name=Mohammed Tanveer Ahmed
user.email=97100223+mdtanveer801@users.noreply.github.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
 ESCOD
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/ssl/certs/ca-bundle.crt
```

./ ../practice What it means: You attempted to execute a directory as if it were an executable script or program. Output (bash: ./: Is a directory): Bash informs you that ./ points to a folder

rather than a file you can run, which causes an error. git config --list What it means: This command lists all active Git configuration settings configured on your local system. Key Outputs Visible:It displays your configured Git username (user.name=Mohammed Tanveer Ahmed) and commit email (user.email=97100223+mdtanveer801@users.noreply.github.com). It shows default system parameters such as init.defaultbranch=master, core formatting rules, and SSL backend settings.



```
mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (master)
$ ssh-keygen -t ed25519 -C "mohdahmed3105@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/mdtan/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/mdtan/.ssh/id_ed25519
Your public key has been saved in /c/Users/mdtan/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:taTVdtAmeNCMdM/jHorFsvTVQeF5vOWztTZBQqH7E2g mohdahmed3105@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|          .o*+=.+.|
|           oo*+B .|
|           +ooo=*+|
|          = +oo +X|
|         S +Eo.+=+|
|           ..*.+o.+|
|            o oo.. |
|                .  |
|                   |
+----[SHA256]-----+
```

ssh-keygen -t ed25519 -c "mohdahmed3105@gmail.com"

What it means: This command generates a new Secure Shell (SSH) key pair using the Ed25519 algorithm, adding your email address as a comment for identification.

Output Details:It shows that the private key (id_ed25519) and public key (id_ed25519.pub) were successfully generated and saved inside the C:/Users/mdtan/.ssh/ directory.

It displays the key's unique SHA256 fingerprint and randomart image to visually verify the key.

```
mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (master)
$ cat /c/Users/mdtan/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHvDCX6yjUBr+paNhbW5G4m7zf+aepdRoOR9nhKMIygP mohdahmed3105@gmail.com

mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (master)
$ clip < ~/.ssh/id_ed25519.pub

mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (master)
$ ssh -T git@github.com
Hi mdtanveer801! You've successfully authenticated, but GitHub does not provide shell access.

mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (master)
$ git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'origin'

mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (master)
$ git status
On branch master
```

cat /c/Users/mdtan/.ssh/id_ed25519.pub

Use: Prints the contents of your public SSH key file directly inside the terminal.

Output: Displays the full Ed25519 public key string and your email address. clip < ~/.ssh/id_ed25519.pub

Use: Copies the contents of your public SSH key directly into your Windows clipboard so you can paste it into GitHub.

ssh -T git@github.com

Use: Tests your SSH connection to verify that GitHub recognizes your key.

Output: Displays the success message: "Hi mdtanveer801! You've successfully authenticated, but GitHub does not provide shell access." git push -u origin main

Use: Attempts to upload your local commits to the remote main branch.

Error Output: Returns error: src refspec main does not match any because the local branch is still named master, meaning the main branch doesn't exist locally yet.

git status Use: Checks the current state of the repository. Output: Confirms that you are currently working On branch master.

```
mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   Bash Scripting 2.pdf
        new file:   "Self-Introduction\342\200\223Experienced-Candidate.pdf"
        new file:   abcd.docx


mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (master)
$ get remote add origin git@ ssh -T git@github.com
bash: get: command not found

mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (master)
$ git remote add origin git@ssh -T git@github.com
error: unknown switch `T'
usage: git remote add [<options>] <name> <url>

    -f, --fetch           fetch the remote branches
    --tags                import all tags and associated objects when fetching
                          or do not fetch any tag at all (--no-tags)
    -t, --track <branch>  branch(es) to track
    -m, --master <branch>
                          master branch
    --mirror[=(push|fetch)]
```

git status

Use: Checks the current state of your repository.

Output: Shows you are on branch master with files staged and ready to be committed, but no commits made yet.

git remote add origin git@ ssh -T git@github.com

Use: A mistyped command attempt.

Error Output (bash: get: command not found): Bash fails because get is not a valid terminal command (it should be git).

git remote add origin git@ssh -T git@github.com

Use: An attempt to link a remote repository, combining the git remote add command with the SSH connection-testing command (ssh -T) by mistake.

Error Output (error: unknown switch 'T'): Git throws an error because -T is not a valid option for git remote add, displaying the proper usage guide for the command.

```
mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (master)
$ git remote -v

mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (master)
$ git remote add origin git@github.com:mdtanveer801/Git_practice.git

mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (master)
$ git remote -v
origin  git@github.com:mdtanveer801/Git_practice.git (fetch)
origin  git@github.com:mdtanveer801/Git_practice.git (push)

mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (master)
$ git branch -M main

mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (main)
$ git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'github.com:mdtanveer801/Git_practice.git'

mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (main)
$ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
```

git remote -v (First instance)

Use: Lists current remote repositories; initially showed no configured remotes.git remote add origin git@github.com:mdtanveer801/Git_practice.git

Use: Successfully links your local repository to your remote GitHub repository (Git_practice.git) using the name origin.

git remote -v (Second instance) Use: Verifies the remote connection.

Output: Displays the configured origin URL for both fetch and push.

git branch -M main Use: Renames your local working branch from master to main.

git push -u origin main

Use: Attempts to push your staged files to the remote main branch.

Error Output (error: src refspec main does not match any): Fails because you have not yet run git commit on your local repository, meaning there are no commits on the main branch to push yet. git status

Use: Checks repository status. Output: Confirms you are on branch main with changes staged, but lists "No commits yet".

```
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   Bash SCripting 2.pdf
        new file:   "Self-Introduction\342\200\223Experienced-Candidate.pdf"
        new file:   abcd.docx


mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (main)
$ git commit -m "practicegit1"
[main (root-commit) ccc9ba4] practicegit1
 3 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Bash SCripting 2.pdf
 create mode 100644 "Self-Introduction\342\200\223Experienced-Candidate.pdf"
 create mode 100644 abcd.docx

mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (main)
$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 602.53 KiB | 2.56 MiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:mdtanveer801/Git_practice.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

mdtan@DESKTOP-T7V22P3 MINGW64 ~/OneDrive/Desktop/practice (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.
```

git commit -m "practicegit1"

Use: Permanently saves your staged files (Bash SCripting 2.pdf, the resume PDF, and abcd.docx) into your local repository history as the initial root commit with the message "practicegit1".

Output: Confirms the creation of the root commit (ccc9ba4) across 3 changed files.

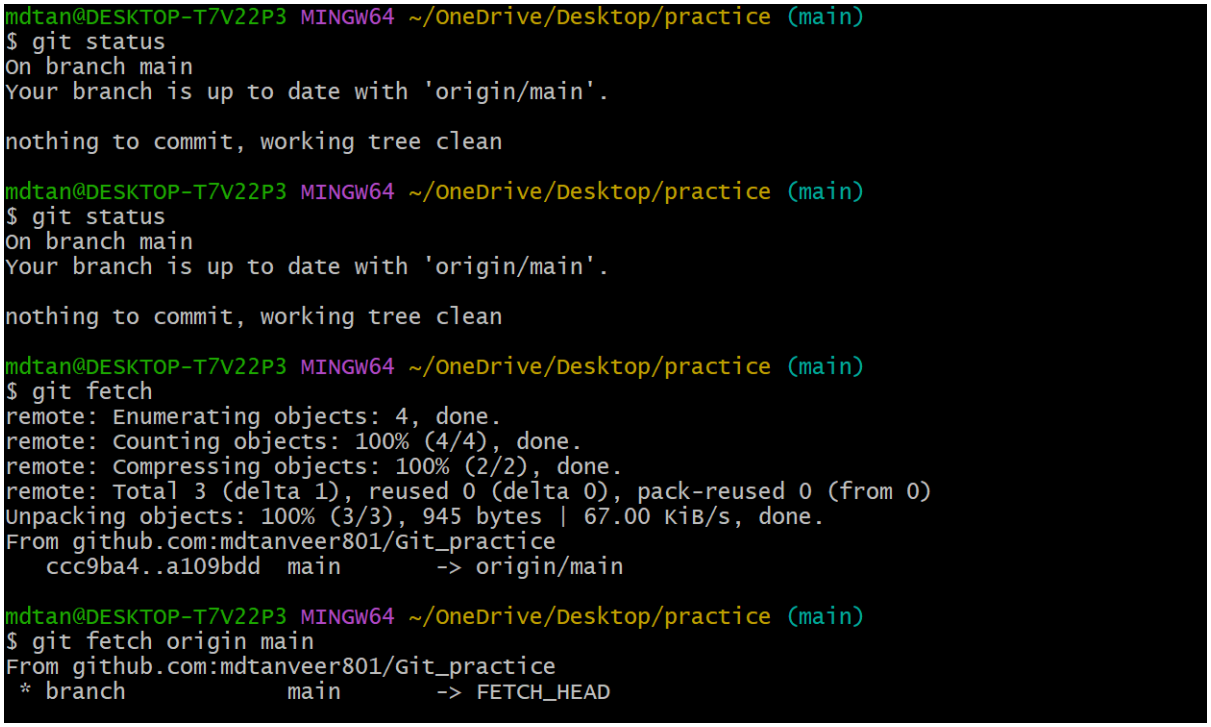
git push -u origin main

Use: Successfully uploads your local commits to the remote GitHub repository (Git_practice.git) on the main branch, and sets up tracking so future commands can simply use git push or git pull.

Output: Displays object enumeration, compression, and confirmation that the main branch was successfully linked and pushed to origin/main. git status

Use: Checks the repository status.

Output: Confirms that you are currently on branch main and your working tree is fully synchronized and up to date with origin/main.



git status (Executed twice)

Use: Checks the current state of your working directory.

Output: Confirms that you are on branch main and your working tree is clean with nothing to commit.

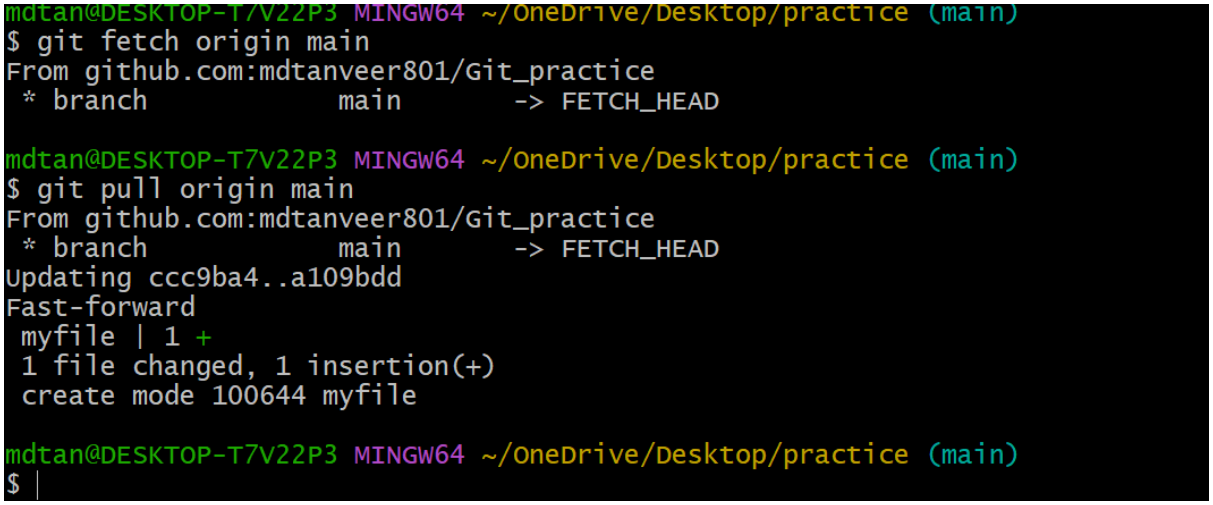
git fetch Use: Downloads the latest changes and history from the remote repository without automatically merging them into your local files.

Output: Shows that new changes were detected on the remote main branch, updating local tracking from commit ccc9ba4 to a109bdd.

git fetch origin main

Use: Explicitly downloads updates specifically for the main branch from the remote repository named origin.

Output: Confirms that the main branch was successfully fetched and recorded into FETCH_HEAD.



git fetch origin main

Use: Downloads the latest updates specifically for the main branch from the remote repository named origin without modifying your local files yet.

Output: Records the fetched branch reference into FETCH_HEAD.

git pull origin main

Use: Combines git fetch and git merge by downloading updates from the remote main branch on origin and immediately merging them into your current local branch.

Output: Performs a fast-forward update from commit ccc9ba4 to a109bdd, showing that myfile was successfully downloaded and added to your local workspace (1 file changed, 1 insertion(+)).